# CSE 260: Discrete Structures in Computer Science

Spring 2027 - Section 001 - 4 credits

| **Instructor** | **Dr. Priya Raman (ramanpri@msu.edu)** |
|---|---|
| Lectures | Tuesday & Thursday, 12:40 - 2:00 PM, 1279 Anthony Hall |
| Recitation | Wednesday, 3:00 - 3:50 PM, 1240 Engineering Building |
| Office hours | Monday 1:00 - 2:30 PM and by appointment, 3132 Engineering Building |
| Course site | D2L (all homework is submitted on D2L) |

## Course description

Discrete mathematics for computer science: propositional and predicate logic, proof techniques, sets, functions, relations, counting, discrete probability, and graphs. The course emphasizes writing clear and correct proofs.

## Grading

| **Component** | **Weight** |
|---|---|
| Homework (10, lowest dropped) | 25% |
| Recitation quizzes (4) | 10% |
| Midterm exams (2) | 40% |
| Final exam | 25% |

Homework is due on D2L by 11:59 PM on the Friday it is listed. Quizzes are given during recitation. Classes start Monday, January 11, 2027 and end Friday, April 30, 2027. There are no classes during spring break, March 8 - 12.

## Attendance

Attendance is expected at every class meeting. If you must miss class for an illness or a university-sanctioned activity, email the instructor before class when possible. Repeated absences will make it difficult to succeed in this course, because much of the material is introduced only in class.

## Academic integrity

Article 2.III.B.2 of the Student Rights and Responsibilities states that the student shares with the faculty the responsibility for maintaining the integrity of scholarship, grades, and professional standards. You may discuss ideas with classmates, but the work you submit must be your own. Using AI tools to write answers you submit is not allowed unless an assignment says otherwise.

## Accommodations

If you have a documented disability and need accommodations, contact the Resource Center for Persons with Disabilities (RCPD) and share your Verified Individualized Services and Accommodations (VISA) form with the instructor within the first two weeks of the semester.

## Late work

Late work is accepted up to 48 hours after the deadline with a 10% penalty per day. Work more than 48 hours late receives no credit unless an extension was arranged in advance. Technology problems are not an excuse, so submit early and keep copies of your files.

## Tentative schedule

| Week | Dates | Topics | Due / assessments |
|---|---|---|---|
| 1 | 1/11 - 1/15 | Logic and proofs | - |
| 2 | 1/18 - 1/22 | Predicates, quantifiers | HW 1 (Fri 1/22) |
| 3 | 1/25 - 1/29 | Sets and functions | HW 2 (Fri 1/29) |
| 4 | 2/1 - 2/5 | Sequences and sums | HW 3 (Fri 2/5)<br>Quiz 1 in recitation (Wed 2/3) |
| 5 | 2/8 - 2/12 | Induction | HW 4 (Fri 2/12) |
| 6 | 2/15 - 2/19 | Strong induction, recursion | MIDTERM 1 in class (Thu 2/18) |
| 7 | 2/22 - 2/26 | Counting | HW 5 (Fri 2/26) |
| 8 | 3/1 - 3/5 | Permutations, combinations | HW 6 (Fri 3/5)<br>Quiz 2 in recitation (Wed 3/3) |
| 9 | 3/8 - 3/12 | SPRING BREAK - no classes | - |
| 10 | 3/15 - 3/19 | Discrete probability | HW 7 (Fri 3/19) |
| 11 | 3/22 - 3/26 | Relations | MIDTERM 2 in class (Thu 3/25) |
| 12 | 3/29 - 4/2 | Graphs | HW 8 (Fri 4/2)<br>Quiz 3 in recitation (Wed 3/31) |
| 13 | 4/5 - 4/9 | Trees | HW 9 (Fri 4/9) |
| 14 | 4/12 - 4/16 | Graph algorithms | - |
| 15 | 4/19 - 4/23 | Boolean algebra | HW 10 (Fri 4/23)<br>Quiz 4 in recitation (Wed 4/21) |
| 16 | 4/26 - 4/30 | Review | - |
| Finals | 5/3 - 5/7 | Final exam (cumulative) | FINAL EXAM Tue 5/4, 10:00 AM - 12:00 PM, 1279 Anthony Hall |